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The Inflammatory Response to Skeletal Muscle Injury

Illuminating Complexities

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Contents

Abstract

Injury of skeletal muscle, and especially mechanically induced damage such as contusion injury, frequently occurs in contact sports, as well as in accidental contact sports, such as hockey and squash. The large variations with regard to injury severity and affected muscle group, as well as non-specificity of reported symptoms, complicate research aimed at finding suitable treatments. Therefore, in order to increase the chances of finding a successful treatment, it is important to understand the underlying mechanisms inherent to this type of skeletal muscle injury and the cellular processes involved in muscle healing following a contusion injury.

Arguably the most important of these processes is inflammation since it is a consistent and lasting response. The inflammatory response is dependent on two factors, namely the extent of actual physical damage and the degree of muscle vascularization at the time of injury. However, long-term anti-inflammatory treatment is not necessarily effective in promoting healing, as

indicated by various studies on NSAID treatment. Because of the factors named earlier, human studies on the inflammatory response to contusion injury are limited, but several experimental animal models have been designed to study muscle damage and regeneration.

The early recovery phase is characterized by the overlapping processes of inflammation and occurrence of secondary damage. Although neutrophil infiltration has been named as a contributor to the latter, no clear evidence exists to support this claim. Macrophages, although forming part of the inflammatory response, have been shown to have a role in recovery, rather than in exacerbating secondary damage. Several probable roles for this cell type in the second phase of recovery, involving resolution processes, have been identified and include the following: (i) phagocytosis to remove cellular debris; (ii) switching from a pro- to anti-inflammatory phenotype in regenerating muscle; (iii) preventing muscle cells from undergoing apoptosis; (iv) releasing factors to promote muscle precursor cell activation and growth; and (v) secretion of cytokines and growth factors to facilitate vascular and muscle fibre repair. These many different roles suggest that a single treatment with one specific target cell population (e.g. neutrophils, macrophages or satellite cells) may not be equally effective in all phases of the post-injury response.

To find the optimal targeted, but time-course-dependent, treatments requires substantial further investigations. However, the techniques currently used to induce mechanical injury vary considerably in terms of invasiveness, tools used to induce injury, muscle group selected for injury and contractile status of the muscle, all of which have an influence on the immune and/or cytokine responses. This makes interpretation of the complex responses more difficult. After our review of the literature, we propose that a standardized non-invasive contusion injury is the ideal model for investigations into the immune responses to mechanical skeletal muscle injury. Despite its suitability as a model, the currently available literature with respect to the inflammatory response to injury using contusion models is largely inadequate.

Therefore, it may be premature to investigate highly targeted therapies, which may ultimately prove more effective in decreasing athlete recovery time than current therapies that are either not phase-specific, or not administered in a phase-specific fashion.

According to published studies in the US,[1-3] contusion (caused by a blunt, non-penetrating object) and strain injuries account for approximately 90% of all sports-related injuries. Contusions are the most frequent type of injury reported by athletes – up to 60% of all reported injuries in the US. The muscle groups of the arms, hands, legs, feet and buttocks are most commonly affected.[4] Apart from the expected mechanical damage to muscle cells at a microstructural level, skeletal muscle contusion injury involves capillary rupture and infiltrative bleeding, oedema and inflammation. These changes can lead to haematoma formation and can cause compartment syndrome in areas where volume expansion is limited by fascial planes.[5,6] This phenomenon is characterized by severe pain. Due to extensive damage to blood vessels and surrounding tissue, oxygen and nutrients are prevented from reaching nerve and muscle cells,[4,5] thereby exacerbating the already existing tissue damage.

Contusion injuries have been clinically documented for a variety of sports in the sports-related literature, but these reports are largely limited to the relative frequency of contusion injury compared with other types of injury in both contact and non-contact sport, such as rugby,[7,8] soccer,[9] wrestling,[10] hurling,[11] luge[12] and long-distance triathlon.[13] Very few studies have focused on other clinically applicable issues, such as diagnostic criteria. One study did compare the efficacy of different diagnostic methods – these included range-of-motion testing, plasma creatine kinase concentration and ultrasonography. Ultrasound results correlated best with the size of the haematoma (measured in this study as an indicator of severity of injury) and was recommended as the best diagnostic tool available.[14] Symptoms of contusion injuries often do not follow a typical pattern, but generally involve soreness, pain with active or passive motion, limited range of motion, or a combination of these.[6,15] Despite occurrence of these debilitating symptoms, the large variability in the severity of injury complicates both the treatment of and research into this condition to such an extent that a universally accepted, evidence-based treatment modality remains to be developed.[5,16] However, before this can be achieved, a better understanding of the mechanisms and the role of players involved in contusion injury is needed.

Interactions between the immune system and skeletal muscle may play a significant role in modulating the course of both the contusion injury and the subsequent muscle repair. As a result of muscle injury and capillary rupture, blood-borne inflammatory cells and cytokines gain direct access to the site of injury.[17-19] The magnitude of the inflammatory response depends on two main factors, namely the severity of injury and the degree of vascularization of the tissue at the time of injury.[19] These factors may be interrelated, with the more severe injury often occurring at sites having a much greater degree of vascularization, leading to a larger inflammatory response. However, this does not discount a role for resident immune cells or non-immune cells' contributions to the inflammatory response. In addition, although it is generally accepted that cytokines (e.g. tumour necrosis factor-α [TNFα], interleukin [IL]-1β and IL-6) are integral to the inflammatory response,[20-23] the exact role of each player is still unclear.

In the early response to skeletal muscle injury, neutrophils are the most abundant immune cells at the injury site, but within the first 24 hours, neutrophil numbers start to decline and the number of macrophages increases.[18,24] Despite this decline in neutrophil numbers, these neutrophils remain functionally active and their numbers are still elevated from baseline at the site of injury for approximately 5 days, after which their activity gradually returns to pre-injury levels. Although the function of neutrophils in response to muscle injury is well described (sections 1.1.2 and 1.1.3), the specific roles played by the macrophages in response to injury *in vivo* are not equally well understood.[18,25]

Studies on muscle injury[18,26,27] and skin wound healing[28,29] are in agreement that limiting the extent of inflammation in the short term, during the 'neutrophil phase' of inflammation, could limit the extent of damage, with the benefit of decreasing pain and swelling. In the sports-related literature, fibrotic scar formation is associated with slow and incomplete recovery of muscle strength.[30,31] Some studies even suggested that fibrosis after an injury may be the cause of recurrent muscle tears.[32,33] In skin wound healing, excessive scar formation is ascribed to an excessive inflammatory response.[28,29] These investigations may provide scientific support for the beneficial effects of acute anti-inflammatory treatment sought after by athletes. However, depletion of macrophages (i.e. the later phase of inflammation) has long been known to have negative consequences on the healing process, including reduced muscle regeneration, satellite cell differentiation and growth of muscle fibres.[34,35] Using small animal models and *in vitro* models (e.g. cell culture), it is possible to assess the effects of existing drugs on the different phases of recovery. NSAIDs are commonly prescribed for contusion injury, and many athletes use over-the-counter NSAIDs over long periods of time to reduce post-exercise pain and swelling.[36-39] Although short-term NSAID treatment during the early repair phase (1–3 days) may result in a modest inhibition of inflammatory symptoms (swelling and pain), it may in fact have negative effects on the healing of the injured muscle if taken for a longer period (in excess of 3 days).[40-43] These may include promotion of fibrosis[40] and inhibition of both the early[42] and later phases of muscle cell regeneration.[41] A rodent study on muscle reloading (which results in muscle inflammation and necrosis) recently reported decreased necrosis and increased ED2$^+$ macrophage infiltration, which is associated with muscle regeneration and repair, when NSAIDs are administered 8 hours prior to insult, but not when administered 8 and 16 hours after.[44] This result obtained in an animal model warrants further investigation, since it suggests that it may be most effective to use NSAIDs as early as possible.

In contrast, steroidal anti-inflammatory agents, such as corticosteroids, have been shown in rodent models to have several adverse effects on healing, such as delaying the clearance of debris at the site of injury and prolonging the muscle regenerative process and recovery of muscle strength, even after only one acute administration.[5,43] Despite the fact that anabolic steroids are inappropriate drugs for athlete use and have detrimental effects on myocardial function,[45] this has not prevented research into possible reasons why athletes abuse drugs in this class. Within the context of attempting to understand how inflammatory responses and muscle healing can be influenced, a rodent study has shown that acute anabolic steroid treatment (the prohibited substance nandralone decanoate in particular) is associated with faster healing and restoration of force development in rodents subjected to experimental contusion injury.[5] In the latter study, steroid treatment was also associated with an initial increase in the inflammatory cell response, suggesting once again that the inflammatory response is a positive force in the healing process.

Together, these findings suggest on the one hand that treatment options for athletes are currently far from optimal, and on the other hand that the different stages to muscle injury and recovery are more complex than previously understood, and that neither can be properly addressed without a greater understanding of the role of the inflammatory response. Both the immune- and muscle-cell responses remain to be fully elucidated, with more detailed attention to the time-course of events, the molecular responses initiated and how these are interrelated, in order for future treatments to target specific processes at specific phases. To achieve this, it is vital to study these phases in a standardized way.

The present article therefore aims to elucidate the importance of studying the immune response to skeletal muscle injury by (i) briefly explaining cellular processes known to occur in skeletal muscle after contusion injury, focusing on the early and the late immune cell responses; (ii) discussing important considerations in choosing an appropriate contusion injury model (e.g. invasive vs non-invasive designs); and (iii) providing a review of results reported in studies focusing on immune parameters (rather than parameters of muscle fibre recovery) and specifically using muscle contusion injury models.

1. Responses to Skeletal Muscle Injury

The response of skeletal muscle to injury follows a fairly consistent broad pattern, irrespective of the underlying cause of injury, e.g. contusion, strain or laceration. Three distinct phases have been identified (for review see Jarvinen et al.[15]), namely destruction, repair and remodelling. However, due to distinct differences in the mechanisms of injury, the details within these phases are likely to differ to a certain extent. In contusion injury, the 'destruction phase' is characterized by the disruption of muscle ultrastructure, as well as extensive damage to the vasculature, which may lead to the formation of a haematoma. Due to this characteristic disruption of the vasculature, the pro-inflammatory immune response is frequently followed by cell death of the damaged muscle fibres. Hurme et al.[46,47] described the time-course of events typically seen after severe contusion injury, induced with a spring-loader hammer. In these immunohistochemical and ultrastructural studies, primary damage has been described in terms of the spreading of unchecked necrosis within the preserved basal lamina of injured fibres, for a distance of 1–2 mm. An early mechanism to limit this damage was reported on day 2 after injury: necrotic fibres were 'closed off' by the formation of a 'demarcation band' – a membrane that demarcated the necrotic part of fibres from non-necrotic parts. Furthermore, in these studies, macrophages were reported to be responsible for mopping up necrotic debris by phagocytosis, phagocytosing necrotic tissue only up to the demarcation membranes.

Following the destruction phase, the 'repair phase' consists of numerous processes, including phagocytosis of the damaged muscle tissue, regeneration of the striated muscle, production of a connective-tissue scar and capillary revascularization. During the third and final 'remodelling phase', the formation of newly regenerated muscle fibres reaches completion and reorganization of the muscle leads to functional repair and a decrease in the size of the lesion.

These three phases are usually closely associated and can overlap (see figure 1, adapted from Li et al.[24]), making temporal resolution of the individual phases difficult. For the purpose of this article, the discussion will be limited to two main parts, both relevant to the immune response following injury: the first part will focus on the inflammatory process and possible contribution of immune cells to secondary damage, while the second will focus on the resolution of muscle injury and the roles of macrophages, cytokines and growth factors in this process.

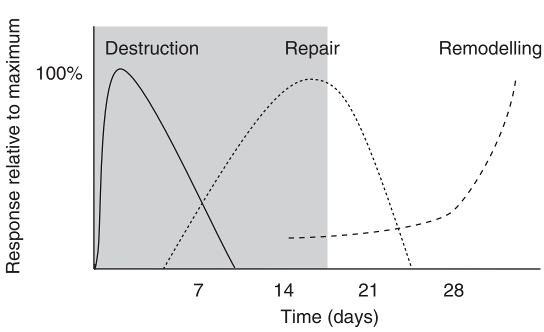

Fig. 1. Illustrative time-course of regeneration of injured muscle includes three chronologically overlapping, but distinct, physiological events: destruction, repair and remodelling (adapted from Li et al.,[24] with permission). The main timeframe for immune cell involvement (see figure 2 for detail) is indicated by shading.

1.1 Inflammation, Leucocyte Infiltration and Secondary Damage

1.1.1 Vascular Involvement

Following a mild contusion injury, the vasculature in the skeletal muscle is usually not disrupted.[20] Therefore, the arterioles within the injured area can dilate, increasing blood flow to the site of injury. This localized vasodilation may be induced *via* two mechanisms. One mechanism for vasodilation is via histamine released from mast cells present within the damaged area.[26] A second effect of this localized histamine release is an increase in capillary permeability at the site of injury via enlargement of the capillary endothelial pores. As a result, an increase in the numbers of phagocytic leucocytes and levels of plasma proteins, both crucial to the inflammatory response,[26] are seen in and around the damaged tissue.[48] Another mechanism for vasodilation is the 'vascular endothelial growth factor – nitric oxide' (VEGF-NO) pathway.[49,50] VEGF may be secreted by fibroblasts,[51] endothelial cells[52] or monocytes/macrophages[51] in response to hypoxia, oxidative stress, growth factors and cytokines,[53] and activates the nitric oxide and nitric oxide synthase pathways to facilitate vasodilation.[49]

Immediately following severe injury, i.e. one in which the vasculature is extensively disrupted, a process of 'damage control' is initiated. Platelets adhere to the exposed collagen, become activated as a result and start to release pro-inflammatory mediators such as serotonin (5-HT), histamine and thromboxane A_2 (TxA_2). Following formation of the platelet plug and the control of haemorrhage, blood-borne immune cells begin to migrate into the area of tissue damage,[15] where they contribute to the localized inflammatory response.

1.1.2 Extravasation

The process of tissue infiltration of circulating, blood-borne neutrophils following injury occurs by rolling, adhesion to and migration through the capillary endothelium.[54] Circulating immune cells roll along the endothelium until interactions between heparin sulphate proteoglycans (HSPGs) expressed on circulating immune cells and endothelium slows them down.[55-57] This is followed by interactions with a variety of cytokines and adhesion molecules,[58,59] which trigger rapid integrin-dependent adhesion to the endothelium.[60]

Most circulating immune cells (lymphocytes, monocytes, eosinophils, basophils and natural killer [NK] cells) express the adhesion molecule $\alpha_4\beta_1$ integrin, which facilitates cell adhesion to the activated endothelium by binding to vascular cell adhesion molecule (VCAM)-1.[61,62] However, neutrophils do not express this integrin.[63] Therefore, neutrophils cannot use this to interact with endothelial cells and instead express β_2 (CD18) integrins. These integrins facilitate endothelial interaction by binding to immunoglobulin-like proteins (e.g. intercellular cell adhesion molecules [ICAMs]), which are also present on the endothelial cell surface.[64] This pathway requires prior activation of the ICAMs by stimuli such as complement components, cytokines and soluble proteins, e.g. fibrinogen and clotting factor X. A number of recent reviews discuss this process of extravasation in more detail.[60,65] At this early stage, the origin of the cytokines is not known, although the resident macrophages, the endothelial cells, the muscle cells themselves and the circulating immune cells are likely to contribute.

The mechanism of neutrophils' migration once they have entered the extravascular space has not been fully elucidated. Healing of dermal wounds has been extensively studied and the proteins that facilitate neutrophil migration in that context include Mac-1 (CD11b/CD18), which is also a β_2 integrin, that was recently shown to facilitate neutrophil migration through monolayers of human synovial and dermal fibroblasts *in vitro*.[66] In this study, ICAM-1 (expressed by fibroblasts) was only involved in this process in the presence of TNFα and/or interferon-γ (IFNγ), again indicating the importance of prior activation of ICAM-1. Recent evidence indicates that skeletal muscle cells also express ICAM-1, albeit not proven in the context of muscle injury.[67] Therefore, the possibility exists that neutrophils may be able to migrate through various tissue types, including skeletal muscle, using a mechanism similar to that used for transendothelial migration. Apart from their expression on cell surfaces, ICAM-1 and VCAM-1 are also found in soluble forms in plasma. Increased concentrations of soluble adhesion molecules are thought to mediate atherosclerotic inflammation.[68,69] In patients with chronic diseases, exercise training has been shown to decrease levels of soluble ICAM-1.[68] However, in the sports-related literature, no significant change in the concentrations of soluble cell adhesion molecules in plasma was reported after acute exercise bouts.[23,69] Together, these studies suggest that while assessment of soluble cell adhesion molecules are sensitive indicators of the beneficial effects of exercise on systemic inflammation, they may not be useful indicators of the effect of exercise stress on immune function in otherwise healthy individuals.

A chronological illustration of white blood cell involvement in the response to skeletal muscle injury is presented in figure 2. Immediately following the injury, neutrophils are the predominant cell infiltrate. Neutrophil infiltration can be detected in the damaged muscle within the first hour following injury. As recently

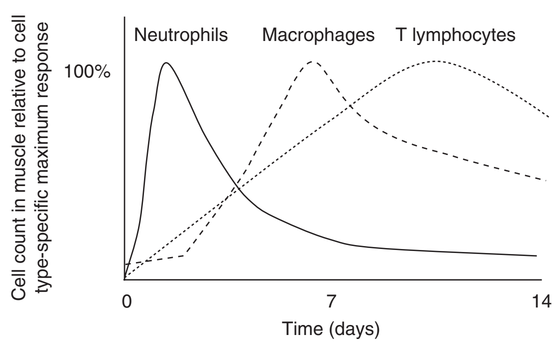

Fig. 2. Chronological involvement of peripheral immune cells following a contusion injury. Cell counts are represented relative to a typical maximum response. The duration that elevated cell numbers apparent in injured tissues may vary depending on the extent of tissue trauma.

reviewed by Tidball,[18] neutrophil cell numbers peak approximately 24–48 hours following injury, and can remain elevated for up to 5 days. The dominant behaviour of the neutrophils – relative to other immune cells – is due to the transfer into the blood of large numbers of preformed neutrophils from the bone marrow, and also an increase in production of new neutrophils in the bone marrow, stimulated by the release of chemical mediators from the inflamed region.[18,48,70] These chemical mediators include complement components (e.g. C5), prostaglandins and leukotrienes, as well as factors released by activated platelets (TxA_2, 5-HT and histamine).[70]

Neutrophils contribute to the post-injury events in two ways: (i) the invading neutrophils have a phagocytic function,[71] clearing the lesion of necrotic debris; and (ii) they magnify the inflammatory process via the release of pro-inflammatory cytokines such as IL-6 and TNFα (see section 1.2.2).[18,72,73] Although phagocytosis and the neutrophil respiratory burst are important mechanisms in the early phase inflammatory response to muscle injury, they can damage the injured muscle even further, and may result in damage to the healthy surrounding tissue. The processes by which this damage occurs are somewhat controversial (see section 1.1.3).

1.1.3 Secondary Damage

Secondary damage in injured tissue may be the result of a variety of cellular and biochemical processes activated in response to the primary insult. Although the extent of this damage may be related to the severity of the primary damage, it does not seem to be dependent on the type of injury.

To date, clear experimental data do not exist to show a direct role for neutrophils in the repair process (for a review, see Tidball[18]). However, neutrophils are the immune cell type that predominates in the injured tissue at the time when secondary damage occurs.[26,46,47] This raises the question of their possible involvement in increasing secondary damage. On the one hand, neutrophils have been shown to generate free radicals during phagocytosis,[74] and neutrophil depletion prior to ischaemia-reperfusion injury was shown to reduce muscle damage by almost 40% in both cardiac[75] and skeletal muscle,[76] providing proof of neutrophil involvement in secondary damage. Similarly, blocking of the neutrophil respiratory burst (using an anti-CD11b antibody) is known to reduce myofibre damage.[77] On the other hand, other phagocytic mechanisms may also come into play: lysosomal enzymes, the proteases responsible for digesting phagocytized matter, are usually contained in specialized membrane-enclosed vacuoles. However, it has been proposed that 'accidental' release of lysosomes from dead or dying cells into extracellular spaces may occur. Also, lysosomes may release their contents within their own cells, thereby causing necrosis.[19] Lysosomal proteolytic pathways were reported to be responsible for at least 40% of muscle atrophy seen after experimentally induced crush injury, which at first seems to argue against immune cell involvement in secondary damage. However, subsequent experiments indicated that these activated lysosomes were associated with macrophages of a phagocytic phenotype within the areas that contain necrotic fibres,[78,79] while lysosomes in muscle and other uninjured tissue were not activated.[80] A recent paper further showed that macrophages recruited by injured muscle were of a phagocytic, pro-inflammatory phenotype, which later converted to an anti-inflammatory phenotype releasing growth factors.[81] It is therefore clear that macrophages do not contribute to secondary damage, but rather have a role to facilitate recovery (roles for macrophages in facilitating recovery are discussed in more detail in section 1.2.1).

From these data, it is clear that there may be multiple contributors to tissue injury. This suggests that a single therapeutic target is unlikely to be sufficient as global treatment for all types and phases of contusion injury.

1.2 Resolution of Muscle Injury

Initiation of injury resolution has previously been defined as the point in time following the injury when the number of neutrophils in the region of damage begins to decrease.[15] However, resolution of muscle damage is multi-faceted, and includes many processes involved in three main phases: resolution of inflammation, angiogenesis and the repair of muscle tissue itself. For the purpose of this article, we will focus on the role of the immune system, as well as related cytokine/chemokine systems, in these different events.

1.2.1 Additional Roles for Macrophages

Although phagocytic macrophages appear to play a major role in the removal of cellular debris, it has been proposed that this is not the only role for macrophages following injury.[73,78,79,82] Unlike neutrophils, macrophages may be divided into subtypes with clear differences in cell surface marker expression. These subtypes are commonly labelled according to their occurrence in different tissue types, e.g. ED1+ (most monocytes and macrophages), ED2+ ('resident' macrophages mainly seen in tissue) and ED3+ (macrophages usually confined to lymphoid tissue).[79,83-85] After muscle injury, these subtypes appear at the injury site at different time-points,[73] suggesting multiple functions for macrophages. A further complexity in teasing out the role(s) of macrophages that was briefly mentioned in section 1.1.3, are the recent suggestions and *in vitro* proof that macrophages can change from one subtype to another, in a manner dependent on their microenvironment. For example, macrophages were reported to express CD163 (ED2) in response to IL-10 and glucocorticoids *in vitro*.[86,87] These claims were substantiated recently by an *in vivo* tracer study, which illustrated a switch in macrophage phenotypes in regenerating muscle.[81] Results from co-culture experiments led these authors to conclude that this subtype switch may be triggered by the actual process of phagocytosis,[81] suggesting yet another role for macrophages in recovery. These multiple functions remain to be elucidated, but it is clear that treatments that target all macrophages may inhibit those with potentially beneficial effects.

Nevertheless, there is consensus in the literature that in injured muscle, ED2+ and ED3+ macrophages appear later than ED1+ cells.[73,79] Since they are rarely observed in degenerating muscle fibres, these two cell types are not considered to be important role players in the phagocytic process.[79] Although research into their exact contribution to the healing process may still pose more questions than answers, a number of studies reported beneficial roles for these macrophages in both damaged fibres and regenerating muscle.[67,82,88-90]

Various models of injury have established at least two mechanisms for cell death in damaged myocytes, namely necrosis and apoptosis.[47,91,92] The occurrence of necrosis as a result of skeletal muscle injury has been well established[46,47,93] and was described in section 1. Although activation of pro-apoptotic signals in skeletal muscle has been reported in failing human cardiac muscle[91] and in skeletal muscle after burn injury,[91] mention of apoptosis in studies describing mechanically injured skeletal muscle is strikingly absent. The reason for the absence of histological evidence for apoptosis may be related to another likely role for macrophages, namely the prevention/limitation of apoptosis.[67,89] For example, satellite cells have been shown to use macrophages as support to promote muscle growth and assist muscle precursor cells to 'escape' apoptosis,[67,89] evident to a greater extent in more differentiated myoblasts,[94] via at least four different adhesion molecule systems that are expressed by macrophages *in vitro* and *in vivo*.[94]

With regard to angiogenesis, it is a well established fact that VEGF is one of the major role players in this process.[50,52,53,95,96] Different sources of VEGF have been identified. For example, differentiating myogenic cells were shown to express VEGF *in vitro*,[52] which was associated with *in vivo* neoangiogenesis in muscular dystrophy.[52] Similarly, skeletal myoblasts expressing VEGF were associated with capillary formation after ischaemia/reperfusion injury in rat heart tissue.[96] Recently, CC-chemokine receptor 2 (CCR2) knockout mice were reported to show delayed angiogenesis and VEGF production during skeletal muscle regeneration.[97] Since CCR2 is an important determinant of monocyte/macrophage recruitment to sites of injury, it is likely that macrophages have an important role in angiogenesis, as suggested by earlier studies.[98-100]

Furthermore, a recent study in mice illustrated that macrophages contribute to muscle regeneration by providing both inflammatory and growth-related mediators.[82] Unfortunately, in this particular study, the study design did not allow for subtype-specific investigations. Cell-type specific information was reported in an *in vitro* study that co-cultured macrophages and myoblasts: $ED2^{+}$ macrophages were shown to be a major contributor to both myoblast proliferation and myotube formation.[101] Furthermore, the latter study also showed that $ED2^{+}$ macrophage-conditioned culture media had similar effects on myoblasts to the macrophages themselves. These results suggest that this subtype of macrophages may serve as a major source of growth factors and cytokines that promote healing.

1.2.2 Inflammatory Cytokine and Growth Factor Involvement

Pro-inflammatory cytokines can be secreted locally in injured muscle by a variety of cell types, including neutrophils, activated macrophages, fibroblasts, endothelial cells and damaged muscle cells (for a detailed review see Cannon and St Pierre[73]). Each of these cell types may well be operating in different time-frames depending on the time-course of their recruitment to the injured area. This complex inflammatory response develops rapidly and only ceases more than a week later (approximately 10–14 days) when the injured area is properly cleared of all damaged tissue.[72,73] The relative contribution of the various cell types present in muscle to this process is unclear and complex to study, since most cytokines originate from more than one cell type and their functions involve several interrelated steps.

One way of elucidating cell type- or cytokine-specific roles would be to inhibit the function of either a specific cell type or a specific cytokine, using antibodies or knockout animals. The eliminated factor may play an independent role or a role in recruitment or activation of another factor. The absence of the eliminated factor may also, in the long-term, lead to compensation by upregulation of other cell types or cytokines. Therefore, such models are complex to interpret. Another state-of-the-art technique (cDNA microarray) allows for investigation into changes in gene expression of inflammatory mediators following muscle injury.[102,103] Although these studies have an important place in science, gene expression does not always correlate with protein expression (i.e. actual production of the inflammatory mediator). Another, less drastic way may be to consider the chronology of cell availability within the damaged area, together with the role of different factors secreted by these cells, to give an indication of the importance of specific cell types to the various phases of the inflammatory process. For example, neutrophils are generally accepted to play a major role in the early inflammatory response to contusion injury, and have been shown to play a role in the recruitment of circulating macrophages to the site of injury, via secretion of chemotactic factors (e.g. IL-1, IL-8).[15,24] However, the increase in macrophage numbers seen in damaged muscle tissue at approximately 2 days post-injury, coincides with the decline in neutrophil numbers.[18] This may on the one hand suggest that macrophages can play a role in sustaining the inflammatory response initiated by damaged cells and neutrophils, at a time when both damaged cell tissue and spent neutrophils are cleared from the injury site. On the other hand, the micro-environment may have changed dramatically in terms of availability of growth factors, etc. by the time of peak macrophage involvement. Therefore, factors secreted by macrophages may have a different net effect compared with similar factors when secreted by neutrophils due to differences in the availability of enhancing or inhibiting factors. Proof of such interactions may be deducted from the information in table II, where the same cytokine (e.g. IL-6) is suggested to have opposing functions at different times during the response to injury.

Tables I and II summarize the chronology of major cell types present in injured muscle, the

Table I. Involvement of the main cell types responding at different stages following contusion injury, with their main secretory products

Cell type	Cytokine/growth factor secreted in different stages of muscle repair		
	destruction of muscle fibres	inflammation	regeneration
Damaged muscle fibres	TNFα IL-1β IL-6 FGF-2	TNFα IL-1β IL-6 MIP-1α MCP-1	IL-1β MCP-1
Endothelial cells	IL-1α IL-1β IL-6	IL-1β IL-6 IL-8 G-CSF M-CSF	IGF-1 HGF FGF PDGF VEGF
Fibroblasts	IL-1α	IL-6 IL-8 G-CSF M-CSF VEGF	
Neutrophils	TNFα IL-1β	TNFα IL-1β IL-8	IL-1β TGF-β
NK cells	IL-1α	IFNγ	IFNγ TNFβ
ED2$^+$ macrophages		TNFα IL-6 VEGF	TGF-β IL-1β IL-15 FGF-2 IGF-1 LIF
T lymphocytes			IFNγ TNFβ IL-1β IL-2 IL-6 TGF-β MIF
B lymphocytes			TGF-β IL-2 IL-6

Continued next page

Table I. Contd

Cell type	Cytokine/growth factor secreted in different stages of muscle repair		
	destruction of muscle fibres	inflammation	regeneration
Muscle precursor cells/myotubes/ regenerating fibres		IL-1β	IL-1β
		IL-6	IL-6
			LIF
			IGF-1, IGF-2
			FGF-1, FGF-2
			TGF-β
			HGF
			MIF
			HMGP
			CNTF

CNTF = ciliary neurotrophic factor; **ED2**$^{+}$ = 'resident' macrophages mainly seen in tissue; **FGF** = fibroblast growth factor; **G-CSF** = granulocyte colony stimulating factor; **HGF** = hepatocyte growth factor; **HMGP** = high-mobility group proteins; **IFN** = interferon; **IGF** = insulin-like growth factor; **IL** = interleukin; **LIF** = leukaemia inhibitory factor; **MCP** = monocyte chemotactic protein; **M-CSF** = macrophage colony stimulating factor; **MIF** = macrophage migration inhibitory factor; **MIP** = macrophage inflammatory protein; **NK** = natural killer; **PDGF** = platelet-derived growth factor; **TGF** = transforming growth factor; **TNF** = tumour necrosis factor; **VEGF** = vascular endothelial growth factor.

different inflammatory cytokines and growth factors produced by these cells and their possible involvement in post-injury events.

Taken together, these findings suggest that besides their role in phagocytosis, immune cells such as macrophages and neutrophils play pivotal roles in mediating muscle repair, either directly by secretion of growth factors, or indirectly by recruitment of other cell types.

2. Injury Models – Technical Considerations

Several confounding factors complicate research on muscle injuries and influence our ability to further elucidate the exact roles of immune cells, particularly macrophages, in damaged tissue. These include inter-individual variations in the severity of injuries that occur in humans, as well as the invasive nature of methods for inflicting injury in some models and methods used for obtaining muscle samples for analysis.

In this section, animal models of muscle injury (invasive and non-invasive) will be described. Particular emphasis will be placed on the relevant technical considerations that may influence results obtained, variability between the different models and the suitability of each for inflammation-related research, all of which contribute to the complexity of studying the immune response to muscle injury.

2.1 Invasiveness of Injury Model

Skeletal muscle contusion injury is a proven result of various methods of inducing mechanical injury in animal models.[20,90,138] The model most commonly described in the literature makes use of a single impact trauma to the muscle, either with or without prior surgical exposure of a selected muscle group.[20,88,90,138-140] For example, the mass-drop injury model introduced by Kvist and colleagues,[141] and also later described by Stratton et al.[142] involves dropping a solid weight with a flat impact surface (varying in diameter and mass) from various heights onto the specific muscle (see figure 3 for a representative illustration). Both localized and systemic inflammation have been studied using the invasive version of this model.[20] Although changes in skeletal muscle expression of the different cytokines (IL-1β, IL-6 and TNFα) were reported, the

Table II. Cytokines and growth factors secreted in response to contusion injury with their possible roles in the recovery of muscle, nerve and capillary tissue

Cytokine/growth factor	Injury-related functions	References
TNFα	Activates proteolysis in the presence of cortisol	104
	Upregulates ICAM-1 and VCAM-1 expression, to facilitate immune cell infiltration	73
	Upregulates IL-1β secretion	105
	Upregulates IL-6 secretion by myoblasts	21
IL-1α	Activates proteolysis independent of cortisol	106
IL-1β	Role to increase protein turnover by inhibition of IGF-1 secretion	74,107
	Upregulates ICAM-1 and VCAM-1 expression, to facilitate immune cell infiltration	73
	Activates endothelial cells to secrete G-CSF and GM-CSF	108
	Upregulates IL-6 production and MCP-1 expression	17,21,24
	Inhibits satellite cell differentiation to enhance proliferation	109
IL-6	Upregulates hypothalamo-pituitary-adrenal axis to increase cortisol levels	110
	Suggested role in rupture of injured myofibres	111
	Upregulates IL-1β secretion	105
	Downregulates TNFα secretion	112,113
	Downregulates concentrations of MIP-2, GM-CSF and IFNγ	113
	Suppresses effect of IGF, resulting in loss of myofibrillar protein	112
	Autocrine secretion by myoblasts associated with increased myoblast proliferation	101
MIP-1α	Recruitment of circulating monocytes into injured tissue	114
MCP-1	Recruitment and activation of macrophages	115,116
	Role in functional restoration of muscle after injury	117
IL-8	Decreases neutrophil infiltration via inhibition of its adhesion to endothelial wall	118
	Upregulates monocyte infiltration into injured tissue	24
G-CSF	Promotes granulocyte production in bone marrow	108
	Decreases satellite cell apoptosis and promotes its proliferation	119
M-CSF	Promotes monocyte/macrophage production in bone marrow	108
IFNγ	Acitvates ICAM-1 to facilitate neutrophil migration from circulation to site of injury	66
	Associated with trauma-induced muscle wasting	120
	Stimulate myoblast proliferation	121
FGF-1, FGF-2	Enhances myoblast proliferation by inhibition of differentiation via inhibition of MyoD expression	109,122,123
IGF-1, IGF-2	Enhances myoblast differentiation by increasing myogenin expression	124
	Stimulates myosin heavy chain production	125
HGF	Enhances myoblast proliferation by inhibition of differentiation via inhibition of MyoD expression, and increases myoblast migration into injured tissue	52,122
PDGF	Enhances myoblast proliferation by inhibition of differentiation via inhibition of MyoD expression, and increases myoblast migration into injured tissue	52,122
VEGF	Facilitates vasodilation in response to ischaemia	50,126
	Promotes angiogenesis after injury	96,97
SDF-1	Role in angiogenesis at sites of inflammation	127
	Provide chemotactic signal to recruit muscle progenitor cells from bone marrow	128

Continued next page

Table II. Contd

Cytokine/growth factor	Injury-related functions	References
TGF-β	Stimulates scar formation	68,129
	Enhances angiogenesis	68
IL-15	Induces myosin heavy chain synthesis, and accelerates process in presence of IGF-1	73
LIF	Suggested function in recovery of intramuscular motor nerves	130,131
	Induces satellite cell proliferation via activation of the JAK2-STAT3 signalling pathway	132
IL-2	Stimulates fibroblast proliferation	133
MIF	Stimulates fibroblast proliferation	133
	Associated with specialization of tissue and inhibition of macrophage infiltration	134
TNFβ	Stimulates fibroblast proliferation	133
HMGP	Maintenance of muscle precursor cell's state of differentiation, with expression decreasing as differentiation progresses	135,136
CNTF	Regeneration of motor neurons in muscle	137

CNTF = ciliary neurotrophic factor; **FGF** = fibroblast growth factor; **G-CSF** = granulocyte colony stimulating factor; **GM-CSF** = granulocyte-macrophage colony-stimulating factor; **HGF** = hepatocyte growth factor; **HMGP** = high-mobility group proteins; **ICAM** = intercellular cell adhesion molecules; **IFN** = interferon; **IGF** = insulin-like growth factor; **IL** = interleukin; **LIF** = leukaemia inhibitory factor; **MCP** = monocyte chemotactic protein; **M-CSF** = macrophage colony stimulating factor; **MIF** = macrophage migration inhibitory factor; **MIP** = macrophage inflammatory protein; **MyoD** = myogenic regulating factor (related to fusion and terminal differentiation of muscle cells); **PDGF** = platelet-derived growth factor; **SDF-1** = stromal cell-derived factor 1 (also known as CXCL12); **JAK2-STAT3** = janus-activated kinase 2 signal transducer and activator transcription 3 protein; **TGF** = transforming growth factor; **TNF** = tumour necrosis factor; **VCAM** = vascular cell adhesion molecule; **VEGF** = vascular endothelial growth factor.

lack of sham-operated groups resulted in failure to account for cytokines released as a result of the surgical procedure itself. Surgical (sham) procedures involving damage to the skin have been used in other areas of research as control, and have been reported to increase circulating cytokine levels, in particular those of TNFα and IL-6, within the first few hours post-surgery.[143] Considering that the inflammatory response to contusion injury is also launched during this same time period, such a confounder could affect results obtained. Therefore, in particular when studying interactions between the immune system and muscle, a non-invasive model of injury is imperative.

One variation of this model uses the placement of a heavier weight on the muscle without impact force (i.e. no impact damage), but for a prolonged period of time, usually two or more hours.[144,145] Apart from this model resulting in more severe damage due to longer-term occlusion of blood flow, it more closely simulates muscle injury incurred in accidents where the subject is trapped, and is therefore not ideally suited for studying sports-related injuries.[146]

Forceps crush injury is another invasive model of contusion injury. Prior to the injury, the muscle is surgically exposed (as with invasive mass-drop), placed in between the jaws of a forceps and then bruising is achieved by pinching the forceps manually or by dropping a weight onto the forceps. Apart from the invasiveness of this method, manual contusion injury is difficult to deliver reproducibly. Despite these complications, this model has been used, with success, to study secondary damage,[147] as well as skeletal muscle regeneration[139,140,148] and possible interventions.[147] However, the inflammatory response has not been investigated.

2.2 Details of Drop-Mass Weight

Murine and rat models have been used to study contusion injury. Given the difference in body size between mice and rats, it is not unexpected that the mass of drop-weights and

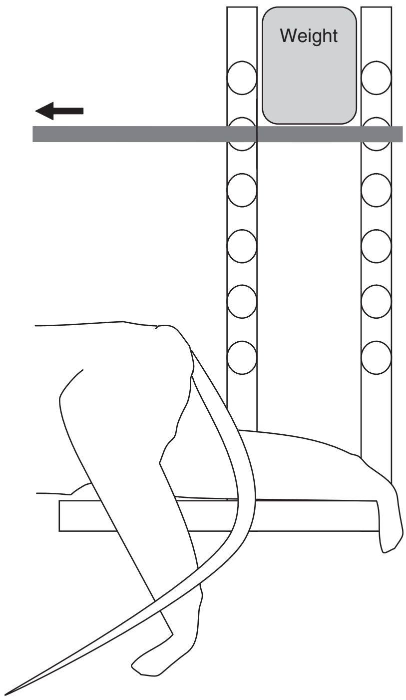

Fig. 3. Representative illustration of a non-invasive, standardized 'drop-mass injury jig'.

hence the impact force used to induce injury also differed. Two specific approaches previously used in rats to induce contusion injury with the mass-drop technique were either relatively heavy weights from a relatively small height (640 and 700 g from 27 and 25 cm, respectively),[142,149-151] or a smaller weight, but from a greater height (171 g from 102 cm).[1,152-154] Similarly, injuries to mice were also typically induced using one of these strategies.[20,88,155] However, other factors in addition to the size of the weight and the drop height also influence the injury produced, for example the shape of the weight.

The impact surface influences both the size of the impact area and severity of the skeletal muscle injury. When a weight with a flat impact surface is used, an injury of uniform severity is achieved, but if a weight with a spherical impact surface is used, the injury induced is more severe in the middle of the impact area compared with the periphery. The infiltration of immune cells into the site of injury was reported to be more concentrated to the middle of the injury when a spherical impactor tip was used than with a flat surface.[5,152,155] The diameter of the weight's impact surface will also influence the impact area and has varied from study to study. If a weight with a larger impact surface hits the muscle belly, the injured area will be wider than when a mass with a smaller impact surface is used. In summary, the characteristics of shape and diameter of the impact surface will both ultimately influence the pattern of injury and the subsequent infiltration of the immune cells. Therefore, in a discussion of existing literature and in future studies, care should be taken to describe the method of injury induction comprehensively. Also, the choices made with regard to the injury induction method will influence the outcome and should be selected (and verified) based on the desired injury to be simulated.

2.3 Selection of Muscle Group

Due to the different metabolic phenotypes seen in different muscle fibres types, the degree of vascularization differs between fibre types. Type I fibres have a high demand for oxygen and show a high degree of vascularization, which provides a rich source of oxygen and nutrients to the fibres. In contrast, type IIb fibres are oxygen independent, with a reduced capillary supply when compared with type I fibres.[156] Although whole muscles most frequently contain a mixture of different fibre types, the proportions in which they are found differ substantially between different muscles and can also differ between different species or individuals within a specific species.[157] To date, the majority of research on skeletal muscle crush injury has made use of the gastrocnemius muscle,[1,5,149-151,153,154,158,159] or the tibialis anterior muscle.[88,140,145,148] Both of these muscles possess type II fibres, but only the gastrocnemius muscle is a mixed muscle, also containing type I fibres.[160] In addition, the gastrocnemius muscle is relatively large, which makes it possible to injure the muscle without bone involvement, which on the one hand is less painful and therefore more ethically acceptable, and on the other making it less likely that the molecular response is complicated by factors released from injured bone or periosteum. The belly-shaped/fusiform nature of gastrocnemius muscle is similar to that of the biceps muscle, which frequently receives contusion injuries in games such as rugby. Therefore, in our opinion, the gastrocnemius muscle is a good candidate to use when studying sports-related contusion injuries. Similarly, for investigation of other sports-related injuries, care should be taken when interpreting results obtained in muscle groups not directly relevant to the sport being studied.

With regard to the immune and cytokine system, it has been shown that muscle fibres secrete cytokines in a fibre-type specific manner. For example, in the triceps, vastus and soleus muscles of humans at rest, TNFα and IL-18 were expressed in type II fibres only, while IL-6 expression was more prominent in type I than in type II.[161] It is presently unclear how the fibre types differ during the phases of muscle destruction, repair and remodelling.

2.4 Contractile Status of the Muscle

The contractile status of the muscle at the time of injury has been shown to influence the muscle's susceptibility to injury. In one study, muscle was shown to be less susceptible to injury when in the maximally contracted state.[152] However, the extent of muscle relaxation was also a factor. Muscles were only fully relaxed after general anaesthesia, highlighting the importance of similar protocols for animal treatment during such experiments.

In sportspersons, injuries occur during practices or competition and will therefore be accompanied by prior exercise and, if not severe, also by subsequent exercise. Exercise, and more specifically contracting muscle, has been shown to contribute to the production of some pro-inflammatory cytokines, such as IL-6.[162,163] However, muscle production of IL-1β and TNFα appears to be independent of contraction.[164-167] Cytokines produced within contracting skeletal muscle cells and released into the circulation, will bring about further downstream cytokine-mediated events, possibly influencing the systems under investigation. However, it is not clear what volumes or intensities of exercise may exacerbate the destruction phase, either delay or promote the repair phase or enhance the remodelling phase. This makes it difficult to provide evidence-based guidelines for athletes' recovery.

Given the above-mentioned considerations regarding muscle injury, 'clean' models currently used to investigate the effect of the immune response to muscle injuries, such as contusion injury models, remain somewhat disappointing. Despite this lack of an 'ideal' model, we will briefly discuss the literature focusing on immune-related investigations using contusion injury models.

3. Immune-Related Studies Using Contusion Models

3.1 Cytokine Focus

Unlike the many studies investigating cytokine and immune responses to stretch-induced injury, studies concerning contusion injury are relatively scarce. Those investigating the immune response following a contusion injury are mostly limited to studying selected cytokines. Nevertheless, the mass-drop injury model in rats has been used by more than one research group to observe acute (hours to a few days) histological changes in the muscle directly after injury and during the recovery phase.[1,149] Parameters that have been investigated include: extent of inflammatory infiltration, the degree of vascular and muscle tissue disruption, collagen formation and tissue re-organization to name but a few. However, these acute observations were limited to the use of light microscopy, with identification of cell types and tissues involved facilitated by using basic stains (uranyl acetate and lead citrate). Changes in muscle strength and biomechanics of the injured muscle were also reported. However, specific investigations with regard to mediators of inflammation or resolution thereof were not included in these studies.

To date, only three papers reported on the involvement of cytokines in muscle injury in a mass-drop contusion injury model by actual measurement of cytokine concentrations or gene expression.[20,151,168] One paper reported transient increases in IL-1β, and TNFα in an invasive mass-drop injury model, with peak concentrations in muscle seen on day 4 post-injury in the group exposed to a mild contusion injury (100 g weight dropped from 130 mm). In the group exposed to a severe contusion injury (200 g weight dropped from 130 mm), IL-1β and TNFα levels rose more gradually, and continued to increase up to day 8 post-injury. IL-6 levels remained unchanged at all timepoints with mild injury, but were relatively high on day 2 post-injury, decreased on day 4 and then increased again on day 8 after severe injury.[20] Unfortunately, no statistical analysis was performed to compare timepoints, so it is not clear whether these changes were significant. Nonetheless, a second study also points indirectly towards a possibly important role for IL-6 and IL-6-related cytokines in regeneration.[151] In a similar mass-drop injury model, the signal transducer and activator transcription 3 protein (STAT3) was exclusively induced in the activated satellite cells, proliferating myoblasts, and surviving myofibres in the regenerating muscle. STAT3 is associated with the leukaemia inhibitory factor (LIF)-signalling molecule, which is an IL-6 super family member, and LIF has been shown previously to play a role in muscle regeneration using other injury models.[130,169]

The third study did not investigate inflammatory cytokines, but rather focused on the inhibitory effect of therapeutic ultrasound on mechano-growth factor expression after injury, suggesting a possible decreased capacity for regeneration.[168] These findings echo the conclusion of an earlier paper, which suggested that pulsed ultrasound therapy may promote the initial satellite cell proliferation phase, but found no significant beneficial effect on overall morphological aspects of muscle regeneration and revascularization.[170]

The role(s) of the cytokines are much more complex than what is presented in these limited studies, as reflected by the detail in tables I and II, and may act both upstream and downstream of proteolysis, thus playing roles in both the destruction and repair phases.[73]

3.2 Other Possible Treatment Options Investigated

Additional information regarding the inflammatory response after injury could possibly be extrapolated from investigations on the efficacy of various NSAIDs on muscle regeneration following contusion injury, although most of these papers did not include direct assessment of inflammatory parameters.[43,140,159] For example, two studies of rat contusion injury reported no negative effect of five different types of NSAID treatment on long-term muscle regeneration,[140,159] except at lethal doses.[140] A third paper reached the same conclusion with regard to muscle regeneration, but did report decreased neutrophil and macrophage infiltration earlier in the recovery phase (day 2 post-injury) and slower resolution of inflammation in response to 5 days of NSAID treatment.[171]

More direct evidence of immune involvement was reported in an animal model of contusion injury, where cyclo-oxygenase-2 (COX-2) inhibition by NSAID infusion were shown to result in faster restoration of microcirculation disrupted as a result of a mass-drop injury, thereby reducing skeletal muscle secondary tissue damage – measured in this study as leucocyte rolling and adhesion to the vascular endothelium.[90] Although this seems to prove a positive effect of NSAIDs immediately after injury, other studies suggest that total inhibition of the inflammatory phase does not benefit the capacity for regeneration. For example, a study using myogenic precursor cells isolated from COX-2 knockout mice, reported a decreased capacity for fusion of these cells in culture.[41] This result is supported by an *in vivo* study, albeit using a model of freeze injury, which reported decreased myofibre regeneration after long-term treatment with another COX-2 inhibitor.[42] In our opinion, the literature with regard to NSAID treatment therefore holds enough proof that an NSAID does not have a negative effect on muscle recovery from injury if its use is limited to 3 days post-injury, but that longer-term treatment has definite detrimental effects on speed of recovery.

Another possible treatment option with few adverse effects may be cryotherapy. Menth-Chiari et al.[172] investigated the relationship between secondary muscle damage after mass-drop contusion injury and the interactions between leucocytes and endothelial cells, essential in the secondary inflammatory response. Five hours following contusion injury, the injured group had a significantly higher number of rolling and adherent neutrophils than the sham group.[172] Using similar visualization and injury methods, cryotherapy after contusion injury was reported to result in decreased numbers of rolling and adhering leucocytes.[173] In addition, although no direct evidence exists to prove that granulocytes contribute to secondary damage (as discussed in section 1.1.3), a number of studies illustrated direct associations between neutrophil infiltration and secondary muscle damage. For example, local superficial cryotherapy was reported to decrease endothelial dysfunction, resulting in decreased granulocyte infiltration, which was linked to a decrease in myofibre membrane disruption (using desmin staining).[93,174] This study illustrates that the role of neutrophils may be an important determinant of the extent of damage, and warrants further investigation. An additional mechanism for the efficacy of cryotherapy in limiting secondary damage has also been suggested: 5 hours of ice therapy was reported to inhibit the extent of loss of oxidative function (assessed by triphenyltetrazolium chloride reduction) in crush-injured tissue.[147] Therefore, cryotherapy should be further investigated in models of sport-related injuries, as a possible complimentary treatment to at least prevent secondary damage.

3.3 Contusion Injury: the Ideal Model for Injury Research?

Taken together, the results of the studies discussed (in section 3) show that the inflammatory response seen after contusion injury is in many ways similar to those reported after other types of skeletal muscle injury, e.g. stretch injury[77] or laceration,[175] but with the added benefit of having no skin or tendon involvement. Also, a contusion model allows investigation of events following damage to both muscle and vascular tissue, which is not common in other injury models. This illustrates that it is an appropriate model for studying immune-related events occurring after muscle injuries, with the added benefit that it can be standardized easily *in vivo*.

It is clear that research into the inflammatory responses during muscle repair after contusion injury is limited. Also, due to the very specific aims of these investigations, and the invasive nature of exposing the muscle prior to contusion in some of these studies, which may lead to additional secretion of cytokines by these damaged muscle fibres,[15,130] literature on integrated responses of the cytokine and immune systems is lacking. Although models of other types of injury exist to investigate this topic, these models are not without complications. For example, although a standardized model of *in vivo* stretch injury was recently described in rabbits,[77] this – and other – *in vivo* stretch injury models cannot exclude the involvement of tendons in the response to injury. On the one hand, differences in tendon properties (such as compliance) may increase the variability in the extent of muscle injury induced. On the other hand, injury to the tendons themselves may add to the complexity of the results. Similarly, *in vivo* laceration injury will necessarily involve injury to the skin as well, and since keratinocytes are known to secrete several pro-inflammatory cytokines,[176] this model is also open to added complexity. Characterizing a simple *in vivo* model that yields reproducible injuries at different levels of severity is therefore essential to enable research into the interaction of the immune and muscle systems. Furthermore, such a model may assist in determining more clearly the effects of current treatments on the destruction, repair and remodelling phases associated with muscle injury, which could lead to the development of better treatment modalities.

4. Conclusions

Muscle contusion injuries, common in contact sport, have been studied in animal models, and recently particular attention has been paid to skeletal muscle regeneration after injury. Attention has also been focused on various treatment modalities in order to speed up the recovery process as well as to facilitate anti-inflammatory actions. Although it is known that the immune system plays a major role in skeletal muscle regeneration after injury,[18,43,81,89] little detailed research has been done on the immune components of the physiological response to mechanical muscle injuries, so that some conclusions are currently quite generalized or even conflicting. This makes it difficult to provide incontrovertible evidence-based recommendations for treatment. Current treatment modalities may address some, but not all, of the intended benefits because of the complexity and phasic nature of both the injury progression and the healing process.

This review clearly highlights some of the complexities of the injury response, secondary damage, regeneration and repair. The subsequent remodelling events have not been addressed. We conclude that a single treatment is unlikely to optimally improve healing, but rather targeted treatments, administered in a time-dependent manner, need to be developed. To achieve this, the specific roles of the different cytokines and growth factors, as well as the involvement of the various immune cells in the different phases of muscle injury, remain to be properly elucidated.

This article has also pointed out that certain aspects of the different muscle contusion injury models need to be considered in order to promote a reproducible model of muscle contusion injury. We further suggest that future investigations using injury models, should specifically quantify and report the degree and severity of muscle injury achieved, in order to facilitate interpretation of results and placement of conclusions into context with other research. It is only when these factors are controlled that a physiologically relevant model can be established with which new or existing treatments can be tested, which will further improve our evidence base, particularly for the less-well investigated local immune response to contusion injury.

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